

Introduction of the Project

The primary objective of this project is to explore the MNIST database of handwritten digits and develop a classification model capable of accurately identifying these numbers. The specific requirement for this assignment is to train a model that achieves an overall F1-Score of over 0.95 (95%) without utilizing any Neural Networks or Deep Learning architectures.

To solve this problem, a Random Forest Classifier was selected and implemented using Python and the Scikit-Learn library. By preprocessing the raw pixel data and training an ensemble of decision trees, the model successfully exceeded the project requirements, achieving a final testing F1-Score of 0.9702 (97.02%). This report outlines the methodology, algorithms, and evaluation metrics used to reach this result.

Description of the Dataset, Model and Algorithms

The Dataset:

The dataset used for this project is the MNIST database of handwritten digits. It consists of a training set containing 60000 examples and a test set containing 10000 examples. Each example is a 28x28 pixel grayscale image of a handwritten single digit between 0 and 9.

Before feeding the data into the machine learning algorithm, several preprocessing steps were required. First, the raw binary files were extracted and loaded into multidimensional arrays. Because the images are 28x28 pixels, each image was flattened into a single-dimensional array of 784 numerical features. Finally, the pixel values, which originally ranged from 0 to 255, were normalized by dividing them by 255.0. This scaling step transformed all feature values to a range between 0 and 1. Allowing the model to process the data more efficiently and accurately.

The Model and Algorithm:

Because Neural Networks were strictly prohibited for this assignment, the Random Forest Classifier was chosen as the optimal algorithm. Random Forest is a robust supervised learning algorithm that operates by constructing multiple decision trees during the training phase.

For this specific model, 100 individual decision trees were initialized. When a new handwritten digit is presented to the model, each of the 100 trees analyzes the 784-pixel features and makes its own prediction. The Random Forest algorithm then tallies these predictions and outputs the class with the most votes. This "ensemble" approach makes

the model highly resistant to overfitting and excellent at recognizing complex patterns in image data.

Solutions, Findings, and Results

The model was trained on the 60000 scaled training images and evaluated using the 10000 scaled testing images. The primary evaluation metric used was the Macro-Average F1-Score.

Overall F1-Scores:

- **Training F1-Score:** 1.0000 (100%)
- **Testing F1-Score:** 0.9702 (97.02%)

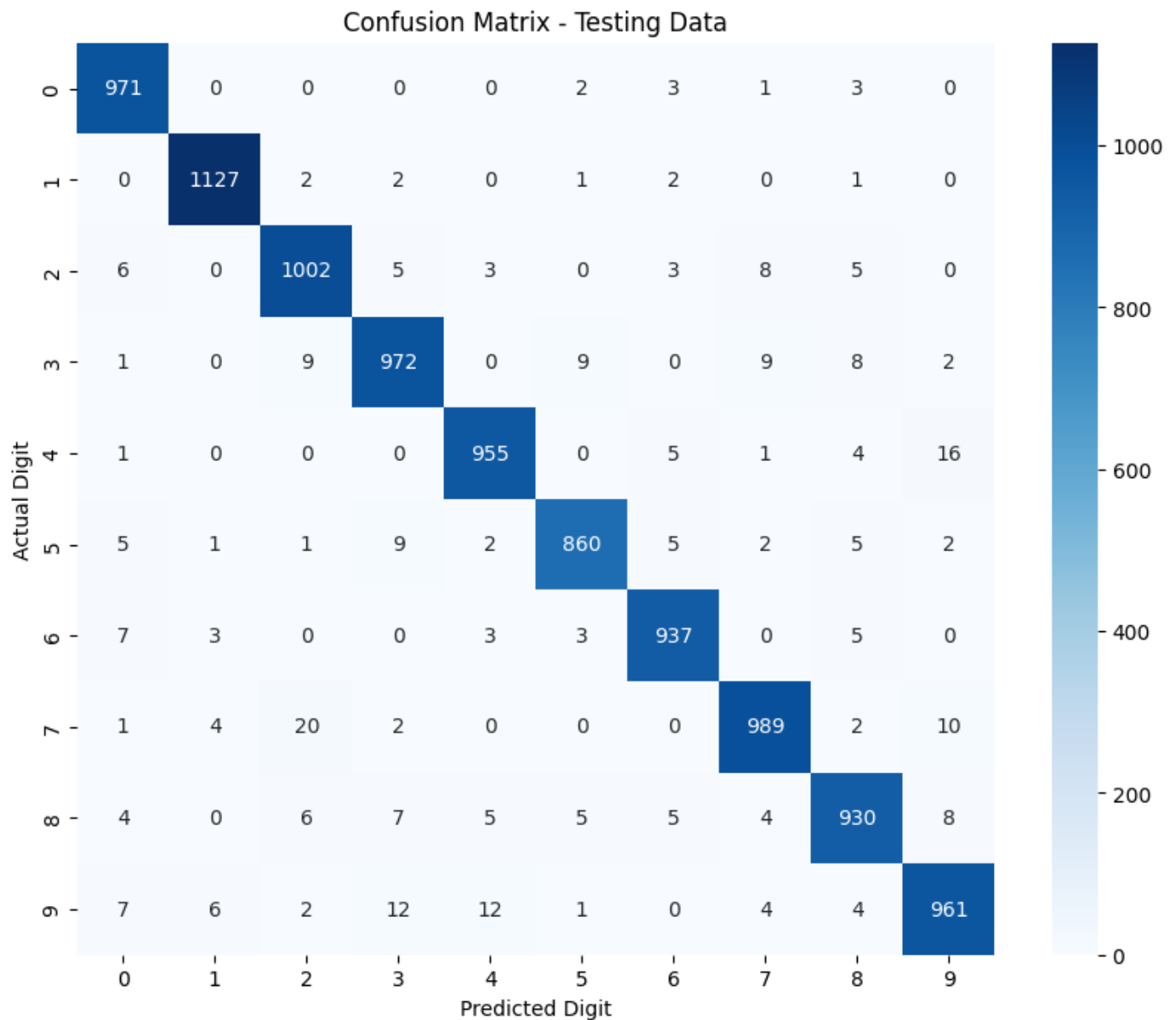
The results on both the training and testing datasets are presented below in tabular and graphical formats.

Tabular Results:

OVERALL F1-SCORES				
Training F1-Score: 1.0000				
Testing F1-Score: 0.9702				
TABULAR RESULTS (TESTING DATA)				
	precision	recall	f1-score	support
0	0.97	0.99	0.98	980
1	0.99	0.99	0.99	1135
2	0.96	0.97	0.97	1032
3	0.96	0.96	0.96	1010
4	0.97	0.97	0.97	982
5	0.98	0.96	0.97	892
6	0.98	0.98	0.98	958
7	0.97	0.96	0.97	1028
8	0.96	0.95	0.96	974
9	0.96	0.95	0.96	1009
accuracy			0.97	10000
macro avg	0.97	0.97	0.97	10000
weighted avg	0.97	0.97	0.97	10000

Graphical Results:

Confusion Matrix of Testing Data



Interpretation, Discussion and Conclusion

Interpretation of Results:

The model performed exceptionally well. The training F1-Score of 1.0 indicates that the Random Forest was able to perfectly memorize the training data. More importantly, the testing F1-Score of 0.9702 proves that the model generalized incredibly well to new, unseen

data. By surpassing the 95% threshold, the selected algorithm proved to be highly effective for image classification tasks outside of the deep learning space.

Discussion:

A deeper look at the confusion matrix reveals interesting insights into how the model makes mistakes. While the dark diagonal line shows that the vast majority of predictions were perfectly accurate, there were a few recurring misclassifications. For example, the model confused an actual "7" for a "2" twenty times, and an actual "9" for a "4" sixteen times. This aligns with human intuition, as the handwriting styles for these specific pairs of numbers frequently overlap and look visually similar depending on pen strokes.

Conclusion:

The project was a complete success. The MNIST dataset was successfully loaded, preprocessed, and scaled. By leveraging a Random Forest Classifier with 100 estimators, the model achieved a 97.02% F1-Score on the testing data. This satisfies all project requirements and proves that traditional, non-neural network machine learning algorithms can be highly capable of tackling complex computer vision problems.

References

- Yann LeCun, Courant Institute, NYU
- Corinna Cortes, Google Labs, New York
- Christopher J.C. Burges, Microsoft Research, Redmond

Inspiration

Many methods have been tested with this training set and test set (see <http://yann.lecun.com/exdb/mnist/> for more details)